Faculty of Computing

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Program:BSCS(Sem 6th)

Course:Compiler Construction

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Lab No 7.

Task 1:

```
#include <iostream>
#include <string>
using namespace std;

string input;
int indexPos = 0;

void S() {
   if(input[indexPos] == 'a') {
      indexPos++;    // match 'a'
      S();          // recursive call
      if(input[indexPos] == 'b') {
         indexPos++;  // match 'b'
      } else {
         cout << "Syntax Error\n";
         exit(0);
      }
   }

}

int main() {
   cout << "Enter string (like ab, aabb, aaabbb): ";
   cin >> input;

   S();

   if(indexPos == input.length())
      cout << "String Accepted\n";
   else
      cout << "String Rejected\n";

   return 0;
}
```

Output:

```
Enter string (like ab, aabb, aaabbb): ab
String Accepted

--------------------------------
Process exited after 46.27 seconds with return value 0
Press any key to continue . . .
```

```
Enter string (like ab, aabb, aaabbb): aabb
String Accepted

--------------------------------
Process exited after 4.62 seconds with return value 0
Press any key to continue . . .
```

```
Enter string (like ab, aabb, aaabbb): aaabbb
String Accepted

--------------------------------
Process exited after 6.1 seconds with return value 0
Press any key to continue . . .
```

Task 2:

```
#include <iostream>
#include <string>
using namespace std;

string input;
int indexPos = 0;

void E();
void Eprime();
void T();
void Tprime();
void F();

void match(char expected) {
   if (input[indexPos] == expected) {
      indexPos++;
```

```
   } else {
      cout << "Syntax Error\n";
      exit(0);
   }
}

void E() {
   T();
   Eprime();
}


void Eprime() {
   if (input[indexPos] == '-') {
      match('-');
      T();
      Eprime();
   }
}
void T() {
   F();
   Tprime();
}


void Tprime() {
   if (input[indexPos] == '*') {
      match('*');
      F();
      Tprime();
   }
}

void F() {

   if (input[indexPos] == 'i' && input[indexPos + 1] == 'd') {
      indexPos += 2;   // match "id"
   }
   else if (input[indexPos] == '(') {
      match('(');
      E();
      match(')');
```

```
    }
    else {
        cout << "Syntax Error\n";
        exit(0);
    }
}

int main() {
    cout << "Enter expression (like id-id*id): ";
    cin >> input;

    E();

    if (indexPos == input.length())
        cout << "String Accepted\n";
    else
        cout << "String Rejected\n";

    return 0;
}
```

Output:

```
Enter expression (like id-id*id): id-id*id
String Accepted

--------------------------------
Process exited after 9.952 seconds with return value 0
Press any key to continue . . .
```

Task 3:

```
#include <iostream>
#include <string>
using namespace std;

string input;
int indexPos = 0;
```

```
void E();
void Eprime();
void T();
void Tprime();
void F();
void match(char expected) {
   if (indexPos < input.length() && input[indexPos] ==
expected) {
      indexPos++;
   } else {
      cout << "Syntax Error\n";
      exit(0);
   }
}
void E() {
   T();
   Eprime();
}
void Eprime() {
   if (indexPos < input.length() && input[indexPos] == '+') {
      match('+');
      T();
      Eprime();
   }
   else if (indexPos < input.length() && input[indexPos] == '-')
{
      match('-');
      T();
      Eprime();
   }
}
void T() {
   F();
   Tprime();
}
void Tprime() {
   if (indexPos < input.length() && input[indexPos] == '*') {
```

```
        match('*');
        F();
        Tprime();
    }
    else if (indexPos < input.length() && input[indexPos] ==
'/') {
        match('/');
        F();
        Tprime();
    }
}
void F() {
    if (indexPos + 1 < input.length() &&
        input[indexPos] == 'i' &&
        input[indexPos + 1] == 'd') {

        indexPos += 2;   // match "id"
    }
    else if (indexPos < input.length() && input[indexPos] == '('
{
        match('(');
        E();
        match(')');
    }
    else {
        cout << "Syntax Error\n";
        exit(0);
    }
}

int main() {
    cout << "Enter expression (like id+id*id, (id+id)*id): ";
    cin >> input;

    E();

    if (indexPos == input.length())
```

```
        cout << "String Accepted\n";
    else
        cout << "String Rejected\n";

    return 0;
}
```

Output:

```
Enter expression (like id+id*id, (id+id)*id): id+id
String Accepted

--------------------------------
Process exited after 13.31 seconds with return value 0
Press any key to continue . . .
```

```
Enter expression (like id+id*id, (id+id)*id): (id+id)*id
String Accepted

--------------------------------
Process exited after 29.11 seconds with return value 0
Press any key to continue . . .
```

```
Enter expression (like id+id*id, (id+id)*id): id*id+id
String Accepted

--------------------------------
Process exited after 13.83 seconds with return value 0
Press any key to continue . . .
```